# Attention Is All You Need

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# The Challenge of Sequential Models

Dominant models like RNNs and LSTMs are inherently sequential, processing data token-by-token.

This sequential nature ($h_t$ depends on $h_{t-1}$) prevents parallelization within examples.

Learning long-range dependencies is a key challenge due to long signal paths and vanishing gradients.

Memory constraints limit batching across examples for long sequences.

# A New Architecture: The Transformer

We propose a network architecture based solely on attention mechanisms.

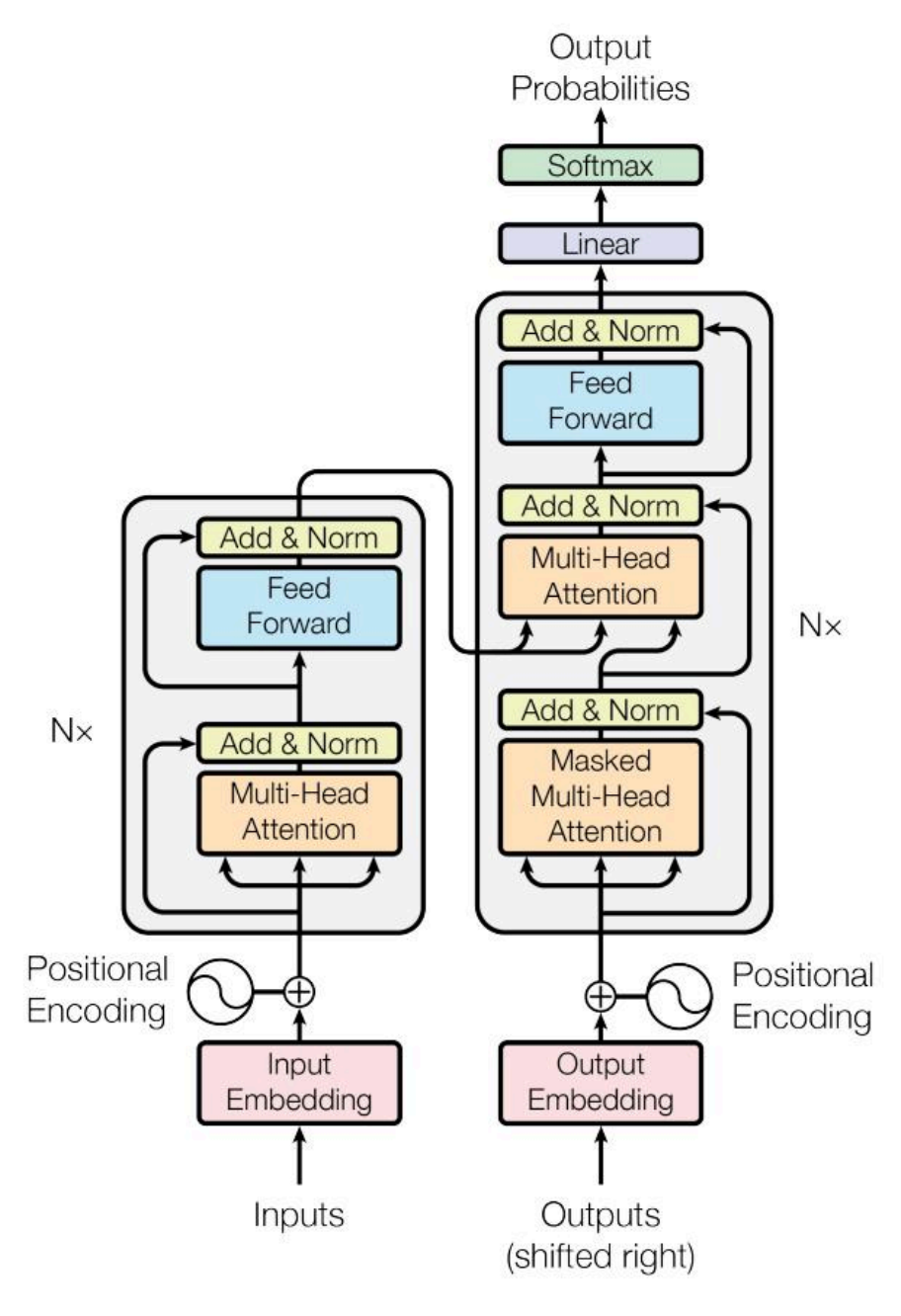
It dispenses with recurrence and convolutions entirely.

This design allows for significantly more parallelization, reducing training time.

Achieves state-of-the-art translation quality while requiring less time to train.

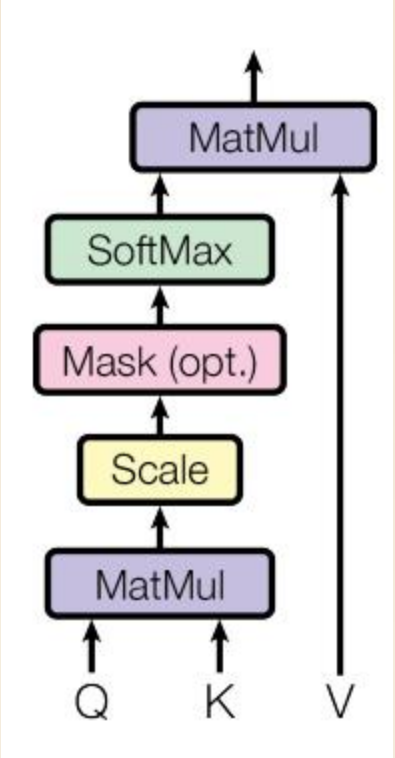
# High-Level Architecture

- The Transformer follows a standard encoder-decoder structure.
- **Encoder:** Maps an input sequence of symbols to a sequence of continuous representations.
- **Decoder:** Generates an output sequence one symbol at a time, consuming previously generated symbols.
- Both components use stacked self-attention and point-wise, fully connected layers.



# The Core Mechanism: Attention

- An attention function maps a query and a set of key-value pairs to an output.
- The output is a weighted sum of the values, where weights are computed based on query-key compatibility.
- We use a specific implementation called Scaled Dot-Product Attention.



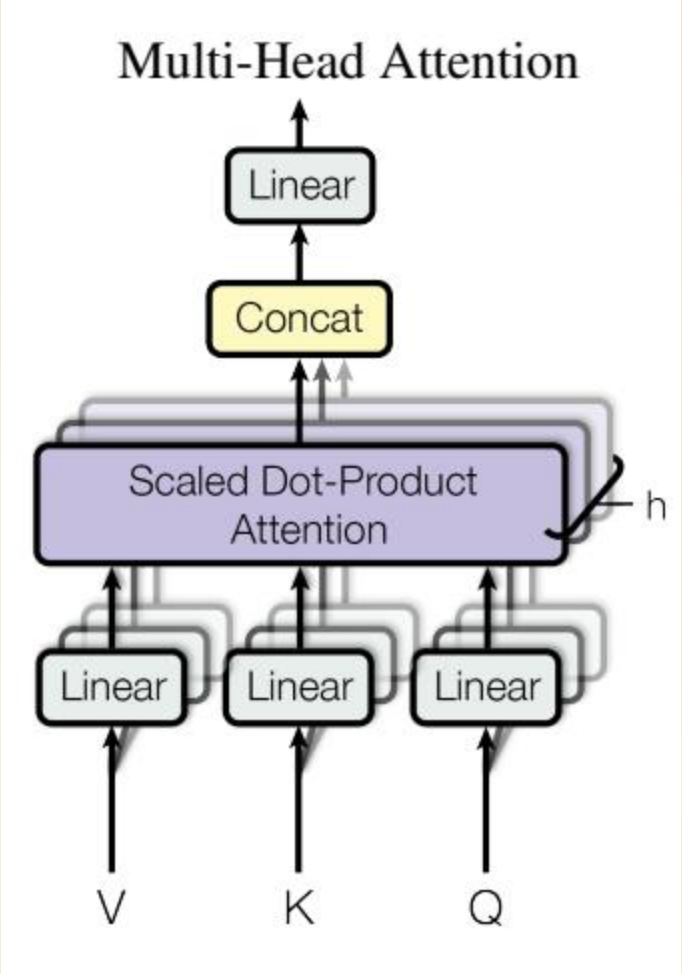
# Scaled Dot-Product Attention

- Input consists of queries (Q), keys (K) of dimension $d_k$, and values (V) of dimension $d_v$.
- Dot products of the query with all keys are computed, then scaled by $\frac{1}{\sqrt{d_k}}$.
- Scaling prevents the softmax function from entering regions with small gradients, stabilizing training.
- A softmax is applied to obtain weights on the values.

$$Attention(Q, K, V) = softmax(\frac{QK^T}{\sqrt{d_k}})V$$

# Multi-Head Attention

- Instead of a single attention function, we use multiple attention 'heads' in parallel.
- Queries, keys, and values are linearly projected $h$ times with different learned projections.
- This allows the model to jointly attend to information from different representation subspaces at different positions.
- Outputs from each head are concatenated and projected to form the final result.



# Multi-Head Attention in Detail

- We use $h = 8$ parallel attention heads.
- For each head, we use reduced key, value, and query dimensions: $d_k = d_v = d_{model}/h = 64$.
- The total computational cost is similar to single-head attention with full dimensionality.

$$\begin{aligned} \text{MultiHead}(Q, K, V) &= \text{Concat}(\text{head}_1, \ldots, \text{head}_h)W^O \\ \text{where head}_i &= \text{Attention}(QW_i^Q, KW_i^K, VW_i^V) \end{aligned}$$

# Layer Structure

### Encoder Layer

- Composed of two sub-layers:

  i. Multi-Head Self-Attention
  ii. Position-wise Feed-Forward Network

- Residual connections and Layer Normalization are applied around each sub-layer.

### Decoder Layer

- Composed of three sub-layers:

  i. Masked Multi-Head Self-Attention
  ii. Encoder-Decoder Attention
  iii. Position-wise Feed-Forward Network

- Masking in the self-attention layer preserves the auto-regressive property.

# Positional Encoding

- Since the model contains no recurrence or convolution, we must inject information about token position.
- We add 'positional encodings' to the input embeddings at the bottom of the encoder and decoder stacks.
- We use sine and cosine functions of different frequencies.

$$PE_{(pos,2i)} = \sin(pos/10000^{2i/d_{\text{model}}})$$
$$PE_{(pos,2i+1)} = \cos(pos/10000^{2i/d_{\text{model}}})$$

# Why Self-Attention?

- Self-attention layers offer advantages in computational complexity, parallelization, and learning long-range dependencies.
- The path length between any two positions is constant ( $O(1)$), unlike in recurrent ($O(n)$) or convolutional ( $O(\log_k(n))$) layers.

| LAYER TYPE | COMPLEXITY PER LAYER | SEQUENTIAL OPERATIONS | MAXIMUM PATH LENGTH |
|---|---|---|---|
| Self-Attention | $O(n^2 \cdot d)$ | $O(1)$ | $O(1)$ |
| Recurrent | $O(n \cdot d^2)$ | $O(n)$ | $O(n)$ |
| Convolutional | $O(k \cdot n \cdot d^2)$ | $O(1)$ | $O(\log_k(n))$ |
| Self-Attention (restricted) | $O(r \cdot n \cdot d)$ | $O(1)$ | $O(n/r)$ |

## State-of-the-Art Machine Translation

- The Transformer outperforms the best previously reported models on WMT 2014 translation tasks.
- The big model establishes a new SOTA BLEU score of 28.4 on English-to-German.
- This performance is achieved at a fraction of the training cost of competing models.

| MODEL | BLEU (EN-DE) | BLEU (EN-FR) | TRAINING COST (FLOPS) |
|---|---|---|---|
| GNMT + RL [38] | 24.6 | 39.92 | $2.3 \times 10^{19}$ |
| ConvS2S [9] | 25.16 | 40.46 | $9.6 \times 10^{18}$ |
| GNMT + RL Ensemble [38] | 26.30 | 41.16 | $1.8 \times 10^{20}$ |
| **Transformer (base)** | **27.3** | **38.1** | **$3.3 \times 10^{18}$** |
| **Transformer (big)** | **28.4** | **41.8** | **$2.3 \times 10^{19}$** |

# Ablation Studies

- We varied components to measure their importance on English-to-German translation.
- (A) 8 attention heads were found to be optimal.
- (D) Dropout is very helpful in avoiding over-fitting.
- (E) Sinusoidal positional encodings perform nearly identically to learned embeddings.

| VARIATION | PPL (DEV) | BLEU (DEV) |
|---|---|---|
| **base (h=8)** | **4.92** | **25.8** |
| (A) h=1 | 5.29 | 24.9 |
| (A) h=16 | 4.91 | 25.8 |
| (D) $P_{drop}$=0.0 | 5.77 | 24.6 |
| (E) Learned Positional Embedding | 4.92 | 25.7 |
| **big** | **4.33** | **26.4** |

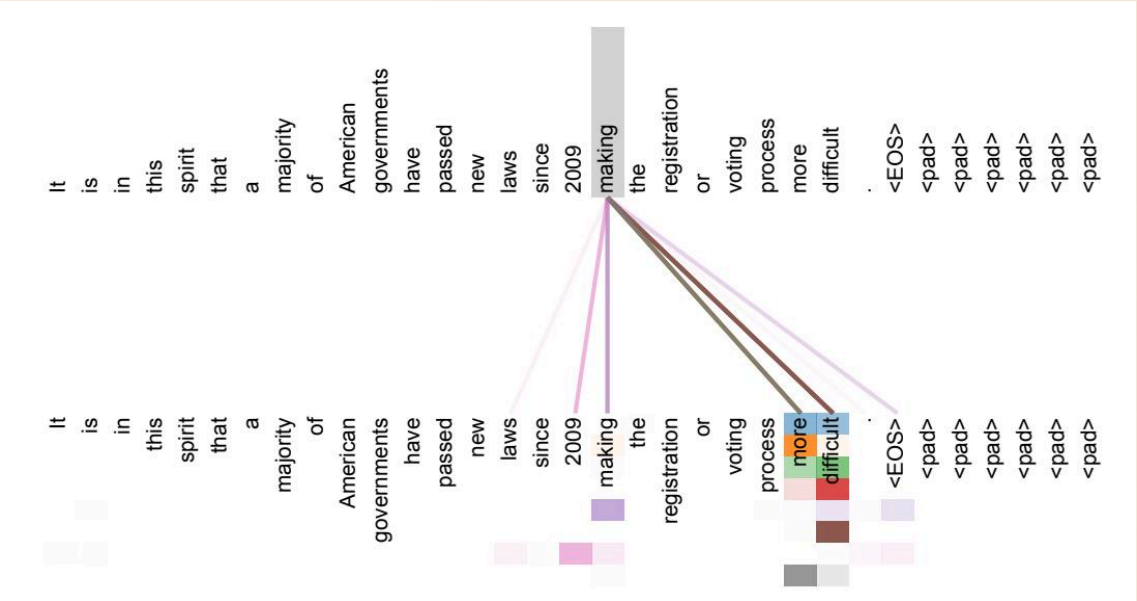
# Generalizing to Other Tasks: Parsing

- To test generalization, we applied the Transformer to English constituency parsing.
- The model performs surprisingly well with minimal task-specific tuning.
- It outperforms the BerkeleyParser even when trained only on the small 40K sentence WSJ dataset.

| PARSER | TRAINING SETTING | WSJ 23 F1 SCORE |
|---|---|---|
| Recurrent Neural Network Grammar | generative | 93.3 |
| Vinyals & Kaiser et al. (2015) | semi-supervised | 92.1 |
| **Transformer (4 layers)** | **semi-supervised** | **92.7** |
| Dyer et al. (2016) | WSJ only | 91.7 |
| **Transformer (4 layers)** | **WSJ only** | **91.3** |

# Visualizing Attention

- Self-attention mechanisms can lead to more interpretable models.
- We can inspect attention distributions to see how the model represents syntactic and semantic structures.
- Different attention heads learn to perform different tasks, such as tracking long-distance dependencies or resolving anaphora.



# Conclusion

- We presented the Transformer, the first sequence transduction model based entirely on attention.
- It trains significantly faster and achieves a new state of the art on machine translation tasks.
- We are excited about the future of attention-based models and plan to apply them to other tasks and modalities.